# Quantum Computation and Communications (EE798V): Problem Set 7

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**Q. 1:** Let the joint probability distribution of two discrete random variables $X$ and $Y$ are given as follows.

| | $X = 1$ | $X = 2$ |
|---|---|---|
| $Y = 10$ | $0.1$ | $0.25$ |
| $Y = 20$ | $0.2$ | $0.2$ |
| $Y = 30$ | $0.1$ | $0.15$ |

(a) Calculate the joint entropy $H(X, Y)$.
(b) Obtain the marginal distribution $p(X)$ and use it to compute the entropy $H(X)$.
(c) Obtain the marginal distribution $p(Y)$ and use it to compute the entropy $H(Y)$.
(d) Obtain the conditional distribution $p(Y|X)$ and use it to compute the conditional entropy $H(Y|X)$.
(e) Obtain the conditional distribution $p(X|Y)$ and use it to compute the conditional entropy $H(X|Y)$.
(f) Verify that $H(X) + H(Y|X) = H(Y) + H(X|Y) = H(X, Y)$.
(g) Calculate the mutual information $I(X; Y)$.

**Q. 2:** If $f : \mathbb{R} \to \mathbb{R}$ is a concave function, then

$$f\left(\sum_{k=1}^{K} p_k x_k\right) \geq \sum_{k=1}^{K} p_k f(x_k)$$

where $x_k$'s are arbitrary real numbers and $\{p_k\}_{k=1}^{K}$ is a valid probability distribution. Moreover, the equality holds if and only if $x_k$'s are all equal. The above relation is known as Jensen's inequality. Note that $\log_2(\cdot)$ is a concave function; therefore, the inequality applies to it.

(a) Let $p$ and $q$ be two probability distributions with the same support set $\mathcal{X}$. Use Jensen's inequality to prove that $D(p||q) \geq 0$. Moreover, the equality holds if and only if $p(x) = q(x), \forall x \in \mathcal{X}$.
(b) If $X$ and $Y$ are two random variables, then show that their mutual information can be expressed as:

$$I(X; Y) = D\left(p(X, Y)||p(X)p(Y)\right)$$

(c) Use 2(a) and 2(b) to conclude that $I(X; Y) \geq 0$ for any two random variables $X$ and $Y$. Moreover, the equality holds if and only if $X$ and $Y$ are independent.
(d) Let $X$ be a random variable with distribution $p_X(\cdot)$ defined on a support set $\mathcal{X}$. Assume that $U$ is a uniform random variable defined on the same support set $\mathcal{X}$. In particular,

$$p_U(x) = 1/|\mathcal{X}|, \ \forall x \in \mathcal{X}$$

Show that the KL divergence $D(p_X||p_U)$ can be expressed as follows.

$$D(p_X||p_U) = -H(X) + \log_2 |\mathcal{X}|$$

(e) Use 2(a) and 2(d) to show that $H(X) \leq \log_2 |\mathcal{X}|$ and the equality holds if and only if $X$ is a uniformly distributed random variable.

**Q. 3:** Let $\rho$ and $\sigma$ be two arbitrary density operators associated with a quantum system described by a Hilbert space $\mathcal{H}$ of dimension $\mathrm{d}$ . Assume that the eigen decompositions of the operators are as follows.

$$\rho = \sum_{i=1}^{\mathrm{d}} \lambda_i |\psi_i\rangle\langle\psi_i| \text{ and } \sigma = \sum_{i=1}^{\mathrm{d}} \mu_i |\phi_i\rangle\langle\phi_i|$$

where $\lambda_i$'s and $\mu_i$'s are (not necessarily distinct) non-negative eigenvalues of $\rho$ and $\sigma$, respectively. Assume, for simplicity, that $\lambda_i, \mu_i > 0, \forall i \in \{1, \cdots, \mathrm{d}\}$. Recall that

$$D(\rho||\sigma) = \mathrm{Tr}\left[\rho\left\{\log\rho - \log\sigma\right\}\right]$$

(a) Use the properties of the density operators $\rho$ and $\sigma$ to demonstrate that $\{\lambda_i\}_{i=1}^{\mathrm{d}}$ and $\{\mu_i\}_{i=1}^{\mathrm{d}}$ are valid probability distributions.

(b) Express $\log \rho$ in terms of $\lambda_i$'s and $|\psi_i\rangle$'s.

(c) Express $\log \sigma$ in terms of $\mu_i$'s and $|\phi_i\rangle$'s.

(d) Show that $D(\rho||\sigma)$ can be expressed as follows.

$$D(\rho||\sigma) = \sum_{i=1}^{\mathrm{d}} \lambda_i \log \lambda_i - \sum_{i=1}^{\mathrm{d}} \sum_{j=1}^{\mathrm{d}} \lambda_i \left(\log \mu_j\right) r_{ij}, \text{ where } r_{ij} = |\langle \psi_i | \phi_j \rangle|^2$$

(e) Show that the set $\{r_{ij}\}_{i=1}^{\mathrm{d}}$ describes a valid probability distribution for each $j \in \{1, \cdots, \mathrm{d}\}$. [Hint: Use the completeness relation.]

(f) Use Jensen's inequality to show that

$$D(\rho||\sigma) \geq \sum_{i=1}^{\mathrm{d}} \lambda_i \log \frac{\lambda_i}{q_i} \text{ where } q_i = \sum_{j=1}^{\mathrm{d}} r_{ij} \mu_j$$

(g) Show that the set $\{q_i\}_{i=1}^{\mathrm{d}}$ describes a valid probability distribution.

(h) Use the non-negativeness of the KL divergence to conclude that $D(\rho||\sigma) \geq 0$.

(i) Use 3(h) to show that the quantum mutual information between two quantum systems $A$ and $B$ is always non-negative.

**Q. 4:** Let $X, Y, Z$ be three random variables. Define the conditional mutual information $I(X;Z|Y)$ as follows.

$$I(X;Z|Y) = H(Z|Y) - H(Z|X,Y)$$

(a) Use the chain rule of entropy to show that

$$I(X;Y,Z) = I(X;Y) + I(X;Z|Y)$$

(b) Similarly show that

$$I(X;Y,Z) = I(X;Z) + I(X;Y|Z)$$

(c) The random variables $X, Y, Z$ are said to form a Markov chain (denoted as $X \to Y \to Z$) if $X, Z$ are conditionally independent given $Y$ i.e., $p(X,Z|Y) = p(X|Z)p(Y|Z)$. If $X \to Y \to Z$, show that the following inequality holds.

$$I(X;Y) \geq I(X;Z)$$

The above inequality is called the data processing inequality.

**Q. 5:** Consider a classical binary symmetric channel (BSC) whose input and output symbols are $X \in \{0,1\}$ and $Y \in \{0,1\}$, respectively. Moreover, the conditional probability matrix $p(Y|X)$ is given as follows.

$$p(Y|X) = \begin{bmatrix} 1-p & p \\ p & 1-p \end{bmatrix}$$

Assume that the input distribution is $p_X(\cdot)$.

(a) Obtain the conditional entropy $H(Y|X)$ as a function of $p$ and show that it is independent of $p_X(\cdot)$.

(b) Use 2(e) to obtain an upper bound of $H(Y)$.

(c) How should $Y$ be distributed to achieve the upper bound?

(d) Does there exist any distribution of $X$ that generates the distribution of $Y$ calculated in 5(c)?

(e) Compute the capacity of the channel $\max_{p_X(\cdot)} I(X;Y)$.